

CarbonNet - AI Carbon Emission

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Abstract- Digital technologies such as cloud platforms, online video streaming, internet browsing, and IoT devices significantly contribute to global carbon emissions, yet traditional carbon calculators often ignore these digital footprints. This project introduces an AI-driven Carbon Emission Monitoring Security System that tracks carbon output from digital activities while ensuring cybersecurity. The system leverages Flutter, Spring Boot, MySQL, Python ML, and a React.js dashboard to monitor activities, detect anomalies, score threats, and generate explainable AI reports. Security features include JWT authentication, MFA, refresh tokens, and RBAC. The AI models automatically retrain with new feedback, providing real-time alerts, analytics dashboards, and secure file scanning.

Keywords – AI Carbon Emission, Digital Carbon Footprint, Machine Learning, Real-time Analytics, Cybersecurity, Carbon Monitoring, Sustainability, Data Privacy, Anomaly Detection, Renewable Integration.

I. INTRODUCTION

Digital technologies such as cloud computing, video streaming, and IoT devices have become integral to modern life, yet their environmental impact is often overlooked. Traditional carbon calculators focus on transportation and manufacturing, neglecting the emissions generated by digital activities. This research introduces an AI-driven system to monitor, analyze, and predict carbon emissions from digital operations, providing real-time analytics and actionable insights for individuals and organizations aiming to reduce their digital carbon footprint.

The AI models support continuous retraining with user feedback, improving prediction accuracy over time. Real-time alerts, analytics dashboards, and exportable reports empower users to understand and reduce their digital carbon footprint. The literature review highlights that digital technologies, especially data centers and cloud infrastructure, already account for about 1% of global energy-related CO₂ emissions, with projections showing rapid growth due to increased digital adoption and AI workloads.

III. PROJECT OBJECTIVES

The main objectives of this research are:

1. To develop a cross-platform application for tracking digital carbon emissions using Flutter and React.js.
2. To implement an AI-powered prediction model using machine learning to forecast emission patterns accurately.
3. To integrate real-time analytics and alerts for immediate feedback on digital carbon output.
4. To ensure system security through JWT authentication,

MFA, and RBAC.

IV. PEER REVIEWED

This research demonstrates a clear understanding of the challenges posed by digital carbon emissions and offers a structured approach to addressing them. The integration of AI, real-time analytics, and robust security features sets the project apart from conventional carbon tracking tools. The methodology is well-organized, and the system design supports practical implementation and scalability.

V. IMPROVEMENT AS PER REVIEWER COMMENTS

Strengths

1. Clear Problem Statement:

The paper effectively highlights the gap in traditional carbon tracking tools, which focus on physical activities and ignore digital carbon emissions. .

2. Methodological Clarity:

The architecture, AI models, and workflow for emission prediction are well-organized and explained in a step-by-step manner, making the system design easy to understand and replicate.

3. Practical Application:

The system demonstrates strong potential for real-world deployment, enabling individuals and organizations to monitor and manage their digital carbon footprint efficiently.

4. Ethical Awareness:

The paper acknowledges the limitations of AI predictions and emphasizes the importance of human oversight and professional consultation for accurate carbon accounting and sustainability planning.

Areas for Improvement

1. Literature Review:

The background section could be strengthened by comparing the proposed system with existing carbon tracking tools and AI solutions, such as Carbon Footprint calculators and digital emission monitors, to highlight unique features and advancements.

2. Dataset Details:

The paper should provide more information about the dataset used for training the AI model, including its size, sources.

3. Ali Mansour, Juman Mohammad, Yury Kravchenko, "Novel Concept-Mining-Based Text Factorization Technique for Effective Feature Extraction"
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VI. CONCLUSION

The AI Carbon Emission Monitoring and Security System successfully bridges the gap between environmental sustainability and cybersecurity. By focusing on digital activities, the system provides transparency, automation, and actionable insights for reducing carbon emissions. The integration of advanced security features ensures data integrity and user privacy, making the solution robust and trustworthy for both individuals and organizations

APPENDIX

1. Tools and Technologies Used: Flutter, Spring Boot, MySQL, Python ML, React.js
2. Security Features: JWT authentication, MFA, RBAC
3. Machine Learning Algorithms: Regression-based predictors, Random Forest ensembles, Neural networks
4. Dataset: Digital activity logs, carbon emission benchmarks

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